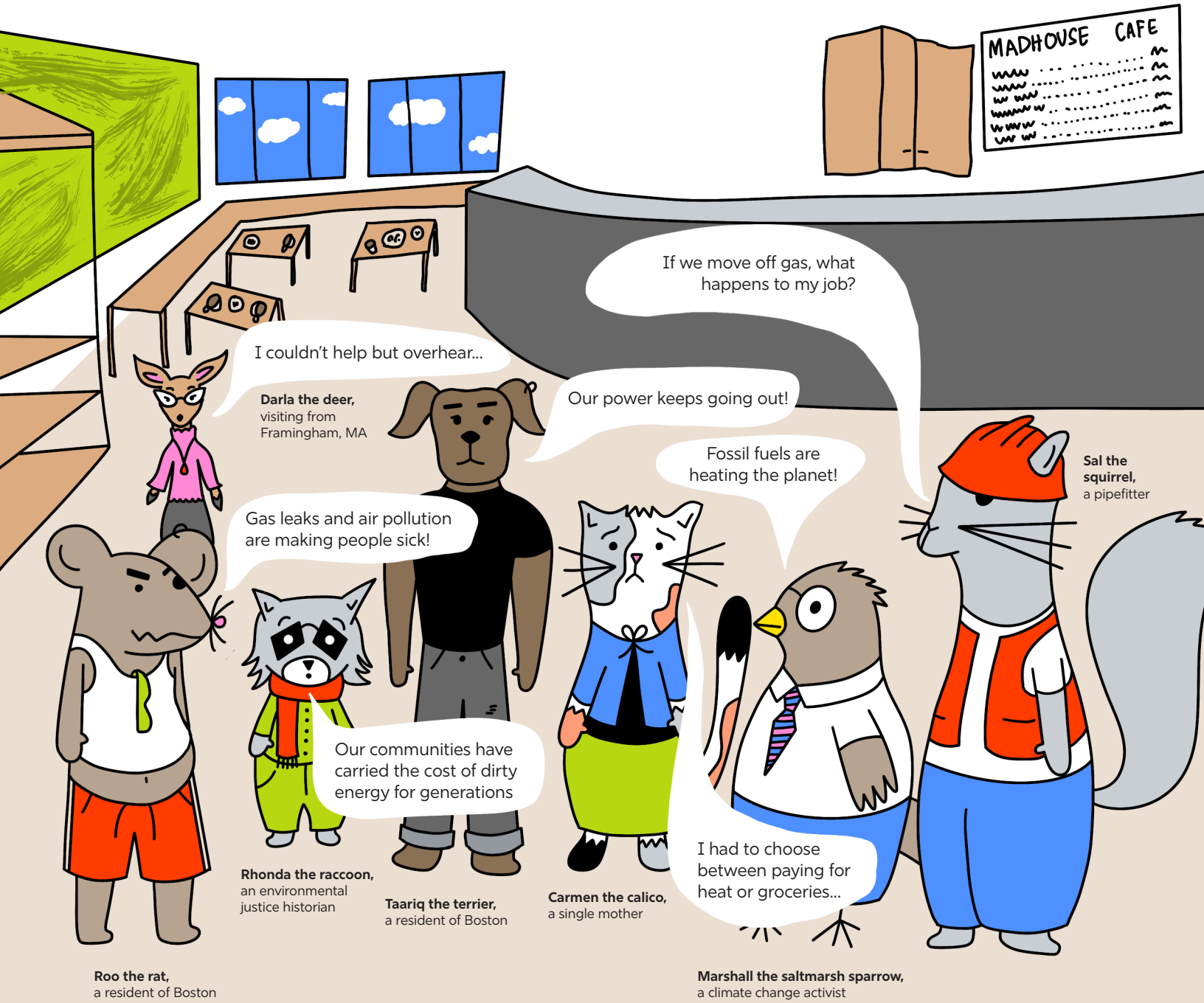
Boston, let's count to TEN!

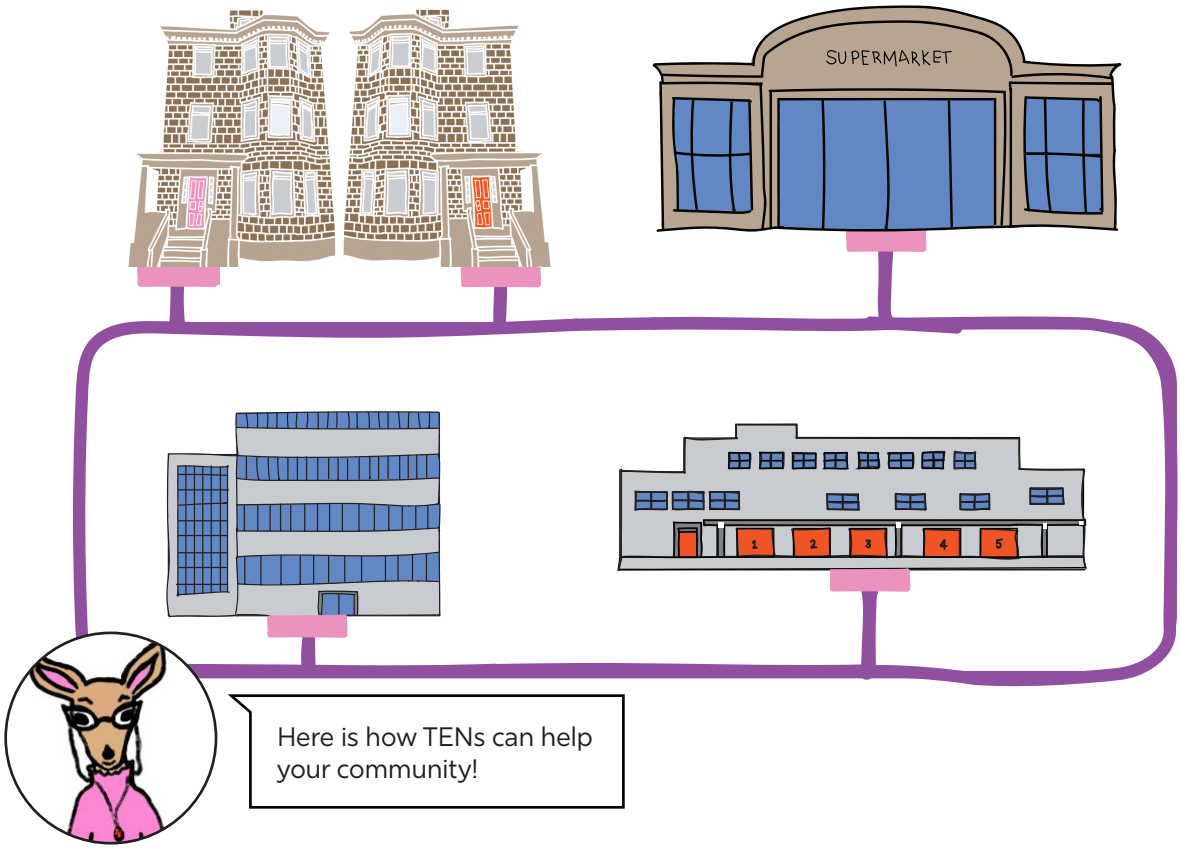
Using thermal energy networks to build an equitable energy future for us all

Bostonians are feeling the strain of an energy system that does not serve our needs. Rising gas and electricity bills make it harder for families to afford a dignified life, aging pipes and grid failures threaten the energy reliability, and neighborhoods long burdened by pollution continue to bear the health and environmental costs of fossil fuels. And as climate change impacts become harder to ignore, it's becoming increasingly urgent for cities everywhere to transition to energy technologies that are clean and just for all.

This toolkit aims to help communities in Boston understand how **thermal energy networks, or TENs,** can offer an alternative solution to the harms of our current energy infrastructure by reducing dependence on gas and improving affordability, reliability, and local air quality.



What is a TEN?



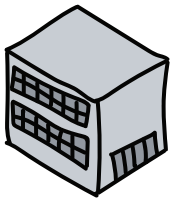
Thermal energy networks, or TENs, like the one recently installed in my neighborhood in Framingham, MA, offer a heating and cooling alternative to traditional gas infrastructure.[1] Instead of using gas to heat homes, thermal energy networks circulate water in a closed loop pipe network, transferring heat between homes, offices, industrial warehouses, and other buildings connected to the network. This water is kept at an ambient temperature by the constant temperature of the earth. The network of pipes will transfer excess heat created by buildings that must be kept cold, like grocery stores, warehouses and factories, ice rinks, and hospitals, and send it through the network to either heat buildings like homes, or be stored underground in boreholes.

Heat pumps installed in each building further cool or heat the water in the network to meet our cooling needs in the summer and our heating needs in the winter. The heat pumps use some electricity to run, but in general TENs reduce strain on our overall electric grid (which as you may have heard is struggling at the moment). You can think of TENs as a way for buildings like refrigerated warehouses or grocery stores in Newmarket to share their excess heat with the surrounding residential neighborhoods instead of just releasing it when their neighbors could use it to get through the winter.

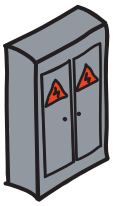
You might be wondering how the TEN installed in my neighborhood impacts me and my neighbors. Aside from being proud to transition away from reliance on harmful fossil fuels, we are comforted by improved indoor air quality, lower rates of asthma for our children, and energy infrastructure that is more resilient to extreme weather events like storms! Workers like my friend Sal can keep their pipefitting jobs because the underground infrastructure of TENs is similar to gas infrastructure and requires the same skillset to install.[2]

What is needed to build a TEN?

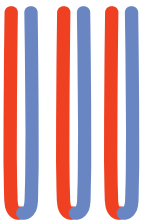
Building a TEN requires certain physical and environmental conditions, as well as coordination across buildings and systems. I can tell you about the key components and site features that make a TEN possible and effective. Let me open my toolkit!



Heat pump
Installed at each building in the network to deliver water at the correct temperature[3]



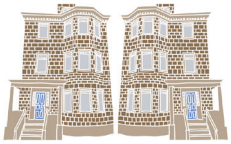
Electrical paneling updates
Many buildings in Boston require electrical paneling updates in order to support heat pumps[6]



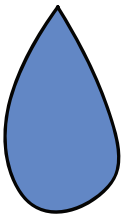
Boreholes
Holes drilled in the ground (between 200 to 700 feet deep and about 6 inches wide) designed to store excess heat produced by buildings[4]

Diverse building uses
Neighborhoods with a balanced mixture of heat sources (buildings that have excess heat, like grocery stores), and heat sinks (buildings that need to be heated, like homes)[2]

Single pipe ambient loop
Pipes similar to gas lines circulate ambient water through a closed loop, allowing each building's heat pump to heat or cool it without wasting water[4]

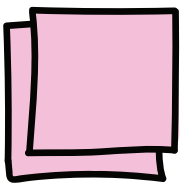


Dense urban environment
Though they can be deployed in many neighborhood types, areas with many closely spaced buildings improve TEN efficiency[2]

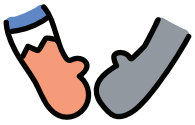


Water
Pipes are filled with water instead of gas. Occasionally, a USDA-approved, non-toxic liquid called propylene glycol is added in order to prevent freezing[4]

Unionized workforce
Skilled gas utility workers transitioning to working on TENs through fair, union-supported jobs will help ensure a just and equitable shift off gas[2]



Building insulation
Well-insulated buildings are great for TENs because heat pumps do not have to work as hard to keep indoor temperatures stable[5]



Community buy in
Broad resident support for fully transitioning to a TEN, ensuring no one is left on the gas system to face rising costs as others switch to clean energy[2]

Will this impact the cost of utilities for my family?

TENs can offer lower and more predictable energy bills. However, depending on the ownership model, residents might need to pay back installation costs over time[2]

Will construction mean that my family will be displaced?

In-home construction can take a few days to a few weeks depending on the age and condition of your building, but it will not mean that your family has to move![7]

Are there community health implications to living near TEN infrastructure?

Instead of gas heating's leaky pipes and toxic furnaces, TENs use durable pipes that circulate water in a closed loop, so they do not impact the surrounding community's health, air or drinking water[2]

How do you ensure that TENs do not harm the earth?

Installing a TEN requires thorough environmental impact reviews and sustainable drilling practices, which, when done diligently, should ensure no harm to the surrounding ecosystem[8]

How do we know if our neighborhood is right for a TEN?

Ideal neighborhoods for TENs are dense and diverse, mixing buildings that emit heat with those that need to be heated[2]

Can TENs help repair harm done to communities by utility companies?

When deployed equitably, TENs can help redirect decision-making power back into communities historically burdened by pollution and disinvestment. This can ensure that the benefits of clean energy flow to those who were the most harmed[2]

Will I lose my job as a utility worker?

Utility workers already have vital knowledge that can be repurposed to install and maintain TENs. That's why TENs are supported by labor justice advocates as a key piece of the just transition[2]

Are TENs more reliable than existing gas infrastructure?

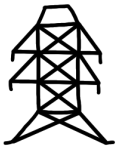
TENs are one of the most resilient energy systems we have. Unlike fossil fuels, thermal energy is plentiful, replenished naturally, and produced much closer to you, so it's more efficient and less prone to outages[2]

What will happen to gas companies like me?

To join a just transition, they must transition to thermal energy companies that serve people over profit, or face strict regulation to ensure health and wellbeing for our communities

What ownership models are possible?

TENs can be developed and managed through a range of ownership structures. Some models have already been realized in communities around the world, while others remain speculative but offer innovative ways to center equity and local control. Let's explore possible ownership models and how they might shape a more just and accountable energy future.



Utility-owned

Operated and maintained by an existing utility company

Eversource is currently running a TEN pilot in Framingham, MA. The network, completed in 2025, serves 125 customer accounts and consists of a mile long loop.[9]

Who bears the cost? Utility companies in the short term, but in the long-term, residents will likely be responsible

Who has a say? Utility companies remain in charge of decision making

Who does this serve? Utility companies primarily serve themselves. While their goals sometimes align with community needs, their goal is profit, not community well-being

City-owned

Owned and managed by a municipal government

The City of West Union, IA funded and owns a TEN. Residents lease the infrastructure to control it. The project was completed in 2014 and serves 12 customers.[10]

Who bears the cost? Cities would bear the upfront costs, likely through grants

Who has a say? Residents would indirectly have a say in public infrastructure. City officials and planners would have a duty to reflect the interests of the public

Who does this serve? As public infrastructure, the TEN would serve residents

Privately-owned

Owned and run by a private company or investor group

In Whisper Valley, TX, the private development company EcoSmart solution developed a TEN that serves more than 800 customer accounts.[11]

Who bears the cost? Private companies cover initial costs but typically recoup them from residents through higher rates

Who has a say? Private companies may engage the community, but residents have little real decision-making power

Who does this serve? Privately owned TENs ultimately serve the company's profit, not the community

Community-owned

Owned and governed by a local cooperative

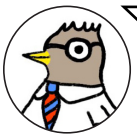
A truly cooperative TEN hasn't been realized yet, but similar models for rooftop solar like the Boston Community Solar Cooperative show it's possible.[12]

Who bears the cost? The community would likely take on initial costs, often through debt

Who has a say? Resident-owners or worker-owners would have a vote in a democratic governance structure

Who does this serve? Ideally, the cooperative serves the community, though it depends on how the governance structure is designed and maintained in reality

What are we asking for?



To create a lasting, people-powered movement, we must build a broad coalition spanning fields, movements, and neighborhoods, cities, and states—and be clear about what we are asking for!

- Clean, reliable, and affordable energy for all. We imagine a future where no household has to choose between paying their energy bills and buying groceries. This means no family will have to spend more than 6% of their income on energy bills and, instead of being burdened by volatile energy costs, energy pricing will remain transparent. When communities get hit with weather events like winter storms, our infrastructure will be poised to handle it and keep communities safe and healthy
- Move Boston, and the U.S. as a whole, away from extractive energy technologies that contribute to global warming, worsened health outcomes, and ecological and economic devastation at home and abroad
- Transition from a profit-based utility system to one that puts people over profit and distributes its benefits equitably across communities
- Build a supported pathway for existing workforce to transition to clean energy jobs. We know that moving away from our reliance on fossil fuels doesn't have to mean current energy sector workers lose their jobs. We see a pathway to a just transition with TENs that allow pipefitters, plumbers, and engineers to use their existing skillsets to aid in the clean energy transition
- Ensure the voices of all community members are heard

Who is doing this work?

Are you interested in creating healthier, more resilient, and more affordable energy sources in your neighborhood? You aren't alone! There are a number of Boston organizations with a powerful history of community organizing for improved health, permanent affordability, and more community say. Here are just a few:

- **Alternatives for Community and Environment:** Fighting to improve indoor air quality and adopt renewable energy sources in Roxbury[13]
- **Dudley Street Neighborhood Initiative:** Building community power to fight back against illegal dumping and create permanent affordability[14]
- **Green Justice Coalition:** Linking labor with environmental justice, and hoping to implement community microgrids throughout Boston[15]
- **Resilient Urban Neighborhoods:** Climate and energy organizations partnering to help local groups implement clean energy[16]
- **Green Roots:** Chelsea and East Boston residents organizing for health and environmental justice[17]
- **Boston Climate Action Network:** Organizing Boston residents to demand urgent action on the climate emergency[18]
- **Boston Ujima Project:** Fighting for collective power, political education, and returning wealth to working class communities of color[19]
- **City Life/Vida Urbana:** Building working class power to fight for racial, social, and economic justice while respecting the environment[20]

How do we talk to the City?

As a resident, worker, or business owner in Newmarket or the surrounding neighborhoods, you might be wondering, what now? We have a few recommendations that can help ensure the transition off gas (and to thermal energy) in Boston is safe, equitable, and affordable for all!

The City recently released a new comprehensive plan for the Newmarket area, and is working on a Climate Action 2030 plan. In PLAN: Newmarket, the City introduces technologies like district and site geothermal as potential energy systems for parts of Newmarket but offers no useful analysis of the area's specific waste heat and thermal loads. Additionally, the city does not explore the potential consequences of implementing TENs on a building-by-building basis. The Plan only considers the boundaries of Newmarket with no regard for opportunities to connect TENs to nearby residential uses.[21]

Given their incomplete analysis of TEN potential in Newmarket, it's important to guide the City in creating plans that are thorough and community oriented. Here are some questions you should ask the City to ensure Boston's energy goals are aligned with the interests of Bostonians:

Will there be legally enforced caps on energy bills at 6% of a household's income to eliminate energy burden for community members?

How will health and affordability for communities most impacted by the harms of historic energy infrastructure be prioritized through the City's energy goals?

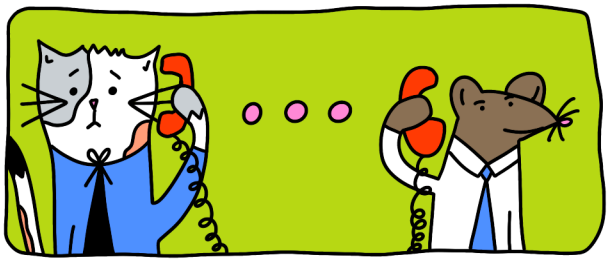
Will the City embed binding language into their plans for implementation of clean energy policies?

How is the City going to address the economic consequences of promoting a transition away from gas on a case-by-case basis instead of a city-wide approach?

How is the City planning to permanently move Boston away from harmful energy technologies instead of just trying to lessen their impacts?

What will legislative protections for workers and workforce development programs look like?

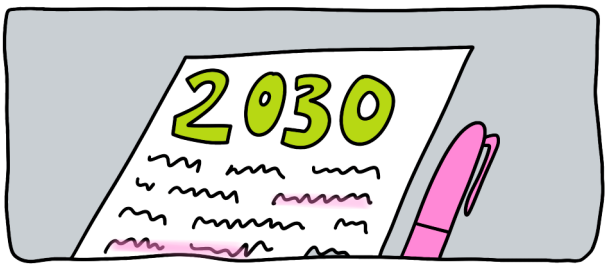
How do we make our voices heard?



Reach out to your city councilor

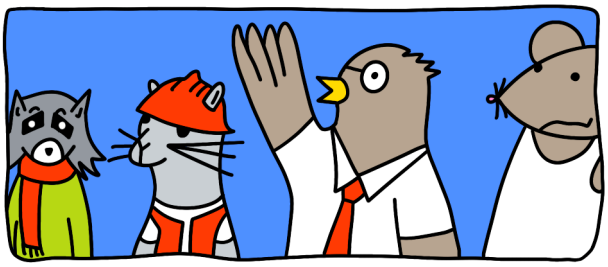
We need your help to call and email City Councilors and demand a just transition! Find contact information for your city councilor at **boston.gov/my-neighborhood**

Make sure to include your full name and District information in your communications with city councilors to ensure your concerns are heard and addressed!



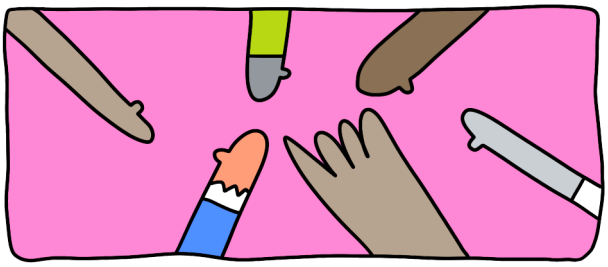
Provide your feedback on plans released by the City

In early 2026, the City of Boston will release the final draft of their 2030 Climate Action Plan and begin welcoming community comments before the final plan is released in Spring 2026. Some portions of the Plan directly address TENs. Read the plan, and be sure to voice your concerns and demands in the feedback survey once it's out!



Show up to community meetings

Find out when energy-related community meetings are happening in Boston, and see if there are opportunities for public comment. Visit **boston.gov/public-notices**



Build community power

Reach out to organizations in Boston that are organizing for energy democracy, worker justice, and community health. There is power in numbers, so talk to the people around you and loudly voice your concerns to help build a movement that cannot be ignored!

Learn more about the energy movement and a just transition:

Climate Justice Alliance
climatejusticealliance.org

Energy Democracy Project
energydemocracy.us

Movement Generation
Just Transition Zine
movementgeneration.org/just-transition

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